Practice Set B

Class 12 Mathematics, Chapter 5: Continuity and Differentiability

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, CBSE SQP 2023-24 — 1 mark

If $f(x) = \begin{cases} \dfrac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases}$ is continuous at $x = 0$, then the value of k is:

(A) -3 **(B)** 0

(C) 3 **(D)** any real number

2 MCQ, NCERT Exemplar Ex 5.3 Q90 — 1 mark

Let $f(x) = |\sin x|$. Then

(A) f is everywhere differentiable

(B) f is everywhere continuous but not differentiable at $x = n\pi$, $n \in \mathbb{Z}$

(C) f is everywhere continuous but not differentiable at $x = (2n+1)\dfrac{\pi}{2}$, $n \in \mathbb{Z}$

(D) none of these

3 MCQ, Board-pattern — 1 mark

The derivative of $2^{\cos^2 x}$ with respect to x is:

(A) $2^{\cos^2 x} \sin 2x \log 2$ **(B)** $-\cos^2 x \cdot 2^{\cos^2 x - 1} \sin 2x$

(C) $-2^{\cos^2 x} \sin 2x \log 2$ **(D)** $-\dfrac{2^{\cos^2 x} \sin 2x}{\log 2}$

4 MCQ, NCERT Exemplar Ex 5.3 Q91 — 1 mark

If $y = \log\left(\dfrac{1-x^2}{1+x^2}\right)$, then $\dfrac{dy}{dx}$ is equal to

(A) $\dfrac{4x^3}{1-x^4}$ **(B)** $\dfrac{-4x}{1-x^4}$ **(C)** $\dfrac{1}{4-x^4}$ **(D)** $\dfrac{-4x^3}{1-x^4}$

5 MCQ, NCERT Exemplar Ex 5.3 Q94 — 1 mark

If $x = t^2$, $y = t^3$, then $\dfrac{d^2y}{dx^2}$ is

(A) $\dfrac{3}{2}$ **(B)** $\dfrac{3}{4t}$ **(C)** $\dfrac{3}{2t}$ **(D)** $\dfrac{3}{4}$

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** The function $f(x)=\begin{cases}\dfrac{x}{\lvert x\rvert+2x^2}, & x\neq 0\\ k, & x=0\end{cases}$ is discontinuous at $x=0$, whatever the value of k. **Reason (R):** $\lim\limits_{x\to 0^-} f(x)\neq\lim\limits_{x\to 0^+} f(x)$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, NCERT Exemplar Ex 5.3 Q43 Differentiate $\tan^{-1}\left(\dfrac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right)$ with respect to x, where $-1<x<1$, $x\neq 0$.	2 marks

Level 2

18 marks

8	Short answer, NCERT Exemplar Ex 5.3 Q12 Find the value of k so that the function $f(x)=\begin{cases}\dfrac{2^{x+2}-16}{4^x-16}, & \text{if } x\neq 2\\ k, & \text{if } x=2\end{cases}$ is continuous at $x=2$.	2 marks
9	Short answer, NCERT Misc. Ex. Q11 Differentiate w.r.t. x: $x^{x^2-3}+(x-3)^{x^2}$, for $x>3$.	3 marks
10	Short answer, NCERT Exemplar Ex 5.3 Q51 If $x=3\sin t-\sin 3t$, $y=3\cos t-\cos 3t$, find $\dfrac{dy}{dx}$ at $t=\dfrac{\pi}{3}$.	2 marks
11	Short answer, NCERT Exemplar Ex 5.3 Q81 If $x=\sin t$ and $y=\sin pt$, prove that $(1-x^2)\dfrac{d^2y}{dx^2}-x\dfrac{dy}{dx}+p^2y=0$.	3 marks
12	Short answer, NCERT Misc. Ex. Q14 If $x\sqrt{1+y}+y\sqrt{1+x}=0$, for $-1<x<1$, prove that $\dfrac{dy}{dx}=-\dfrac{1}{(1+x)^2}$.	3 marks
13	Short answer, NCERT Exemplar Ex 5.3 Q55 Find $\dfrac{dy}{dx}$ when x and y are connected by the relation $\sec(x+y)=xy$.	2 marks
14	Short answer, PYQ-style Find the values of a, b and c for which the function $f(x)=\begin{cases}\dfrac{\sin(a+1)x+\sin x}{x}, & x<0\\ c, & x=0\\ \dfrac{\sqrt{x+bx^2}-\sqrt{x}}{bx^{3/2}}, & x>0\end{cases}$ is continuous at $x=0$.	3 marks

Q No.	Question	Marks
Level 3		**25 marks**
15	Short answer, PYQ-style If $y = \dfrac{\sin^{-1} x}{\sqrt{1-x^2}}$, $-1 < x < 1$, show that $(1-x^2)\dfrac{d^2y}{dx^2} - 3x\dfrac{dy}{dx} - y = 0$.	3 marks
16	Long answer, Board-pattern If $ax^2 + 2hxy + by^2 = 1$, prove that $\dfrac{d^2y}{dx^2} = \dfrac{h^2 - ab}{(hx+by)^3}$.	5 marks
17	Short answer, NCERT Exemplar Ex 5.3 Q61 If $y = (\cos x)^{(\cos x)^{(\cos x)^{\cdots\infty}}}$, show that $\dfrac{dy}{dx} = \dfrac{y^2 \tan x}{y \log \cos x - 1}$.	3 marks
18	Long answer, Board-pattern A function f is defined on $[0,3)$ by $f(x) = \begin{cases} x[x], & 0 \le x < 2 \\ (x-1)x, & 2 \le x < 3 \end{cases}$ where $[x]$ is the greatest integer function. Examine the continuity and differentiability of f at $x = 1$ and at $x = 2$.	5 marks
19	Long answer, Board-pattern If $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t\cos t)$, $0 < t < \dfrac{\pi}{2}$, find $\dfrac{d^2y}{dx^2}$. Hence find its value at $t = \dfrac{\pi}{4}$.	5 marks
20	Case study, Board-pattern **Case Study:** A skate-park designer models the side profile of a ramp. If x is the horizontal distance (in m) and $f(x)$ is the height (in m), the profile is $f(x) = \begin{cases} ax^2 + bx, & 0 \le x \le 1 \\ 2 + \log x, & 1 < x \le e \end{cases}$ For a smooth ride the two pieces must meet without a gap or a sharp corner, so f must be continuous and differentiable at $x = 1$. **(i)** Write the condition on a and b for f to be continuous at $x = 1$. — 1 mark **(ii)** Using (i), find the left-hand derivative and the right-hand derivative of f at $x = 1$. — 1 mark **(iii)** Find the values of a and b, and the height of the ramp at $x = \dfrac{1}{2}$. — 2 marks	4 marks

Answer key

Chapter 5: Continuity and Differentiability, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) -3
2	(B) f is everywhere continuous but not differentiable at $x = n\pi$, $n \in \mathbb{Z}$
3	(C) $-2^{\cos^2 x} \sin 2x \log 2$
4	(B) $\dfrac{-4x}{1-x^4}$
5	(B) $\dfrac{3}{4t}$
6	(A) Both A and R are true and R is the correct explanation of A
7	$\dfrac{dy}{dx} = -\dfrac{x}{\sqrt{1-x^4}}$

Level 2

8	$k = \dfrac{1}{2}$
9	$\dfrac{dy}{dx} = x^{x^2-3}\left(2x\log x + \dfrac{x^2-3}{x}\right) + (x-3)^{x^2}\left(2x\log(x-3) + \dfrac{x^2}{x-3}\right)$
10	$\left(\dfrac{dy}{dx}\right)_{t=\frac{\pi}{3}} = -\dfrac{1}{\sqrt{3}}$
11	$(1-x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + p^2 y = 0$ (proved)
12	$\dfrac{dy}{dx} = -\dfrac{1}{(1+x)^2}$ (proved)
13	$\dfrac{dy}{dx} = \dfrac{y - \sec(x+y)\tan(x+y)}{\sec(x+y)\tan(x+y) - x}$
14	$a = -\dfrac{3}{2}$, $c = \dfrac{1}{2}$, and b is any non-zero real number

Level 3

15	$(1-x^2)\dfrac{d^2y}{dx^2} - 3x\dfrac{dy}{dx} - y = 0$ (proved)
16	$\dfrac{d^2y}{dx^2} = \dfrac{h^2-ab}{(hx+by)^3}$ (proved)
17	$\dfrac{dy}{dx} = \dfrac{y^2\tan x}{y\log\cos x - 1}$ (proved)
18	At $x=1$: discontinuous, so not differentiable. At $x=2$: continuous but not differentiable (LHD $= 1$, RHD $= 3$).
19	$\dfrac{d^2y}{dx^2} = \dfrac{\sec^3 t}{at}$; at $t = \dfrac{\pi}{4}$ it is $\dfrac{8\sqrt{2}}{a\pi}$
20	(i) $a+b=2$; (ii) LHD $= 2a+b$, RHD $= 1$; (iii) $a=-1$, $b=3$, height $= \dfrac{5}{4}$ m.